

# Chapter 25: Geometric Optics

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PHY201  
Lecture 25



# Part I

## The Ray Model of Light

# Outline of Part I: The Ray Model of Light

Introduction to Geometric Optics

Three Models of Light

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Introduction to Geometric Optics

Three Models of Light

# What Is Geometric Optics?

**Geometric optics** models light as **rays** — straight-line paths that carry energy through space.

## The Central Question

- ▶ How do rays propagate through optical systems?
- ▶ How do they change direction at boundaries?
- ▶ How do they form images?

## Fermat's Principle

Light travels along the path of *least time*.

Reflection, refraction, and image formation all emerge naturally from this single

## When Does Geometric Optics Work?

The ray approximation requires:

$$\text{object size} \gg \lambda$$

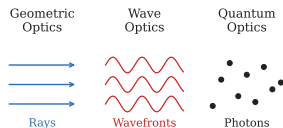
All optical elements must be much larger than the wavelength of light.

# Outline of Part I: The Ray Model of Light

Introduction to Geometric Optics

Three Models of Light

# Three Complementary Models of Light



## Three Regimes

- ▶ **Geometric optics** — rays, image formation
- ▶ **Wave optics** — interference, diffraction
- ▶ **Quantum optics** — photons, transitions

## Not Competing Theories

Each model is a valid *approximation* at a different physical scale.

Camera lens → geometric.  
Soap bubble → wave.

# When the Ray Model Succeeds — and When It Fails

## Model Applicability

| Situation               | Ray Model?  | Why                     |
|-------------------------|-------------|-------------------------|
| Camera lens             | Excellent   | Lens $\gg \lambda$      |
| Shadow behind a doorway | Approximate | Edge diffraction occurs |
| Soap-bubble colors      | Poor        | Interference dominates  |

## The Ray Approximation

In a **uniform** medium, light travels in straight lines.

Direction changes *only* at boundaries between media.

Visible light:  $\lambda \approx 400\text{--}700\text{ nm}$ .

A magnifying glass:  $\sim 5\text{ cm}$ .

**Models are tools, not universal truths.**

Choosing the right model for the right scale is itself a physics skill.

## Check Your Understanding

A magnifying glass forms an image of a small object.  
Which model of light best describes this process?

Come to lecture to see the answer.

## Part II

# Reflection

# Outline of Part II: Reflection

The Law of Reflection

Specular and Diffuse Reflection

# Outline of Part II: Reflection

The Law of Reflection

Specular and Diffuse Reflection

## Light at a Boundary

Whenever light reaches a **boundary** between two media, it simultaneously *reflects* (stays in the original medium) and *refracts* (enters the new medium).

### Organizing Principle

- ▶ **Reflection** governs mirrors
- ▶ **Refraction** governs lenses and prisms
- ▶ **Total internal reflection** is the special case where refraction fails

The normal is a line drawn *perpendicular* to the surface at the point of incidence.

If a ray strikes at  $35^\circ$  from the surface, its angle from the normal is  $90^\circ - 35^\circ = 55^\circ$ .

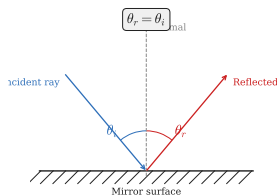
**Important convention:** all angles are measured from the *surface normal*, not from the surface itself.

# The Law of Reflection

## Law of Reflection — (cp2e\_ch25\_s2)

$$\theta_r = \theta_i$$

Both angles measured from the **normal** to the surface.



### Variable Definitions

- ▶  $\theta_i$ : incidence angle (from normal)
- ▶  $\theta_r$ : reflection angle (from normal)
- ▶ All three lines lie in the same plane
- ▶ **Error:** angles from *normal*, not

# Outline of Part II: Reflection

The Law of Reflection

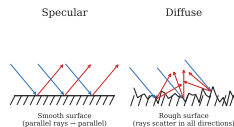
Specular and Diffuse Reflection

# Specular vs. Diffuse Reflection

## Specular Reflection

- ▶ Smooth surface
- ▶ Parallel rays remain parallel after reflection
- ▶ Clear image forms

*Examples:* mirror, calm water, polished metal



## Diffuse Reflection

- ▶ Rough surface
- ▶ Surface normals point in different directions
- ▶ Reflected rays scatter in all directions
- ▶ No image forms

*Examples:* paper, painted wall, wood

Each tiny patch of a rough surface *still obeys* the law of reflection locally.

Diffuse reflection is not a violation — it results from many different local normals

# Image Formation in a Plane Mirror

A plane mirror forms a **virtual image**: rays appear to diverge from a point *behind* the mirror, but no light actually converges there.

## Plane Mirror Image Properties

- ▶ **Virtual** (behind the mirror)
- ▶ **Upright**
- ▶ **Same size** as the object
- ▶ Located the **same distance behind** the mirror as the object is in front

## Real vs. Virtual Images

**Real image:** light rays *actually converge* at that location — can be projected on a screen.

**Virtual image:** rays only *appear* to come from that point when traced backward — cannot be projected.

## Check Your Understanding

A ray strikes a mirror at  $35^\circ$  measured from the *surface*. What is the reflected angle relative to the normal?

Come to lecture to see the answer.

## Check Your Understanding

Why can you see your reflection in a calm lake but not in choppy water?

Come to lecture to see the answer.

## Part III

# Refraction and the Index of Refraction

# Outline of Part III: Refraction and the Index of Refraction

Index of Refraction

Snell's Law

# Outline of Part III: Refraction and the Index of Refraction

Index of Refraction

Snell's Law

# The Index of Refraction

## Index of Refraction $n$ — (cp2e\_ch25\_s3)

$$n = \frac{c}{v}$$

$c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$  (vacuum speed);  $v$  = speed in medium ( $v \leq c$ ).

### Typical Values

|             |             |
|-------------|-------------|
| Vacuum      | $n = 1.000$ |
| Water       | $n = 1.333$ |
| Crown glass | $n = 1.52$  |
| Diamond     | $n = 2.419$ |

### Physical Meaning

$n = 1$ : vacuum (fastest)

$n > 1$ : all real materials

Larger  $n \Rightarrow$  slower  $v$

“Optically denser” = larger  $n$ .

$n$  is dimensionless and always  $\geq 1$  for

# Outline of Part III: Refraction and the Index of Refraction

Index of Refraction

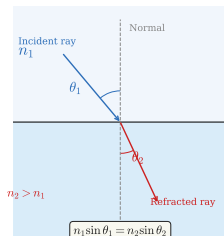
Snell's Law

# Snell's Law of Refraction

## Snell's Law — (cp2e\_ch25\_s3)

At a boundary between media 1 and 2:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$



## Key Variables

- ▶  $n_1, n_2$ : refractive indices
- ▶  $\theta_1$ : incidence angle (from normal)
- ▶  $\theta_2$ : refraction angle (from normal)
- ▶  $n_2 > n_1$ : bends *toward* normal
- ▶  $n_2 < n_1$ : bends *away* from

# Why Does Light Bend at a Boundary?

When a wavefront crosses a boundary at an angle, one side enters the slower medium *first* — that side slows while the other continues at the original speed, tilting the wavefront.

## Wavefront Picture

- ▶ Rays are *perpendicular* to wavefronts
- ▶ Unequal speeds across the wavefront tilt it
- ▶ The tilt of the wavefront *is* the refraction
- ▶ This is why Snell's law is geometrically exact

## Analogy

Think of a car that crosses from pavement onto mud at an angle: the wheels that hit the mud first slow down, turning the car.

Light bends for exactly the same geometric reason.

## Example: Speed of Light in Zircon — (cp2e\_ch25\_ex1)

### Speed of Light in Zircon

Zircon has an index of refraction  $n = 1.923$ . Find the speed of light in zircon.

$$n = \frac{c}{v}$$

$$v = \frac{c}{n}$$

$$v = \frac{3.00 \times 10^8 \frac{\text{m}}{\text{s}}}{1.923}$$

$$v = 1.56 \times 10^8 \frac{\text{m}}{\text{s}}$$

Light travels at roughly *half* its vacuum speed inside zircon.

- ▶ Is  $v < c$ ? ✓
- ▶ Is  $n > 1$ ? ✓
- ▶ Higher  $n \Rightarrow$  slower speed.  
Consistent.

## Example: Identifying a Liquid by Refraction — (cp2e\_ch25\_ex2)

### Identifying a Liquid

Light passes from air ( $n_1 = 1.00$ ) into an unknown liquid. The incident angle is  $\theta_1 = 30.0^\circ$  and the refracted angle is  $\theta_2 = 22.0^\circ$ . Find the index of refraction of the liquid.

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_2 = \frac{n_1 \sin \theta_1}{\sin \theta_2}$$

$$n_2 = \frac{(1.00) \sin 30.0^\circ}{\sin 22.0^\circ}$$

$$n_2 \approx 1.33$$

$n \approx 1.33$  matches **water** in the index table.

The ray bent toward the normal ( $22^\circ < 30^\circ$ ) since  $n_2 > n_1$  — consistent.

- Did the ray bend toward or away from the normal?
- Does  $n_2 > n_1$  predict this? ✓

## Check Your Understanding

When light travels from air into glass, does it speed up or slow down? Does the ray bend toward or away from the normal?

Come to lecture to see the answer.

## Part IV

# Total Internal Reflection

# Outline of Part IV: Total Internal Reflection

The Critical Angle

Applications of TIR

# Outline of Part IV: Total Internal Reflection

The Critical Angle

Applications of TIR

# Total Internal Reflection (TIR)

## Two Conditions for TIR

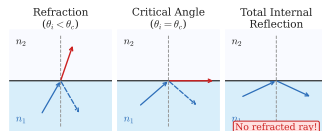
- ▶  $n_1 > n_2$  (denser  $\rightarrow$  less-dense medium)
- ▶  $\theta_i > \theta_c$  (exceeds critical angle)

## Why TIR Occurs

At the critical angle,  $\theta_2 = 90^\circ$  (refracted ray grazes boundary).

Beyond  $\theta_c$ :  $\sin \theta_2 > 1$  is impossible  $\Rightarrow$  all energy reflects.

Going from denser to less-dense medium is required. Light going the other way cannot undergo TIR.



# The Critical Angle

## Critical Angle $\theta_c$ — (cp2e\_ch25\_s4)

Derived from Snell's law with  $\theta_2 = 90^\circ$ :

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

Requires  $n_1 > n_2$  (otherwise TIR is impossible).

### Variable Definitions

- ▶  $\theta_c$ : critical angle
- ▶  $n_1$ : index of the incident medium (larger)
- ▶  $n_2$ : index of the transmitted medium (smaller)

### Reference Values

- ▶ Water–air:  $\theta_c = 48.6^\circ$
- ▶ Glass–air:  $\theta_c \approx 41^\circ$
- ▶ Diamond–air:  $\theta_c \approx 24.4^\circ$

## Example: Critical Angle for Water — (cp2e\_ch25\_ex3)

### Critical Angle: Water to Air

Find the critical angle for light traveling from water ( $n_1 = 1.333$ ) into air ( $n_2 = 1.000$ ).

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

$$\theta_c = \sin^{-1}\left(\frac{1.000}{1.333}\right)$$

$$\theta_c = 48.6^\circ$$

Any ray hitting the water–air surface at more than  $48.6^\circ$  from the normal undergoes TIR.

An observer underwater sees the *entire* above-water world compressed into a cone of  $2 \times 48.6^\circ \approx 97^\circ$ .

► Could TIR occur going from air into water?

# Outline of Part IV: Total Internal Reflection

The Critical Angle

Applications of TIR

# Applications of TIR

## Optical Fibers

- ▶ **Core:** high-index glass ( $n \approx 1.5$ )
- ▶ **Cladding:** slightly lower-index glass
- ▶ Rays hitting core–cladding boundary at  $\theta > \theta_c$  undergo TIR
- ▶ *Uses:* internet, endoscopy, sensing

## Diamond Brilliance

Diamond:  $n = 2.419$ ,  $\theta_c \approx 24.4^\circ$ .

Tiny  $\theta_c$  means rays bounce internally many times before exiting — producing brilliance.

A single optical fiber can carry millions of simultaneous conversations via different wavelengths of light.

Diamond cutters design facet angles specifically to maximize TIR toward the viewer's eye.

## Check Your Understanding

Which of the following transitions can produce total internal reflection?

- (a) Air  $\rightarrow$  glass    (b) Water  $\rightarrow$  diamond    (c) Glass  $\rightarrow$  air  
(d) Vacuum  $\rightarrow$  water

Come to lecture to see the answer.

## Part V

# Dispersion, Prisms, and Rainbows

# Outline of Part V: Dispersion, Prisms, and Rainbows

Dispersion

# Outline of Part V: Dispersion, Prisms, and Rainbows

Dispersion

# Dispersion: Color and the Index of Refraction

**Dispersion** is the variation of the index of refraction with wavelength. Different colors have slightly different values of  $n$ , so they refract by different amounts.

## Color Ordering in Glass

- ▶ **Violet:** shorter  $\lambda$ , *larger*  $n$ , bends more
- ▶ **Red:** longer  $\lambda$ , *smaller*  $n$ , bends less
- ▶ White light separates into a continuous spectrum

In **vacuum**: all colors travel at  $c$  — no dispersion.

In **matter**: different colors travel at different speeds — dispersion occurs.

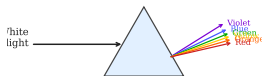
## Crucial Clarification

**Frequency** stays constant at a boundary.

**Wavelength** changes because speed changes:  $\lambda = v/f$ .

# Prisms and Rainbow Formation

Dispersion by a Prism



## Rainbows

**In each droplet:** refraction in → internal reflection → refraction out → color separation.

**Angles (from anti-solar point):**

- ▶ Red:  $\approx 42^\circ$  (outer arc)
- ▶ Violet:  $\approx 40^\circ$  (inner arc)

## Check Your Understanding

Which color of visible light experiences the greatest deflection when passing through a glass prism?

Come to lecture to see the answer.

## Part VI

# Image Formation by Thin Lenses

# Outline of Part VI: Image Formation by Thin Lenses

Types of Lenses

Principal Rays and Image Types

The Thin Lens Equation

# Outline of Part VI: Image Formation by Thin Lenses

Types of Lenses

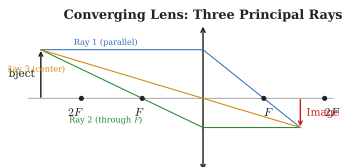
Principal Rays and Image Types

The Thin Lens Equation

# Converging and Diverging Lenses

## Converging Lens ( $f > 0$ )

- ▶ Thicker at the center
- ▶ Parallel rays converge to a real focal point
- ▶ Also called a *convex* lens
- ▶ Applications: camera, eye, projector



## Diverging Lens ( $f < 0$ )

- ▶ Thinner at the center
- ▶ Parallel rays diverge as if from a virtual focal point
- ▶ Also called a *concave* lens
- ▶ Applications: eyeglasses for myopia, viewfinders

Both lens types form images through refraction at curved surfaces.

The *sign* of  $f$  distinguishes converging from diverging.

# Outline of Part VI: Image Formation by Thin Lenses

Types of Lenses

Principal Rays and Image Types

The Thin Lens Equation

# Principal Ray Tracing for a Converging Lens

## Three Principal Rays (use any two)

1. Ray **parallel** to the optical axis  $\rightarrow$  refracts through the **far focal point**  $F$
2. Ray through the **near focal point**  $F \rightarrow$  emerges **parallel** to the axis
3. Ray through the **optical center**  $\rightarrow$  passes through **undeviated**

## Image Location

The image forms where any two principal rays intersect (real image) or appear to diverge from (virtual image).

## Real vs. Virtual

**Real image:** rays actually converge —  
 $d_i > 0$ .

**Virtual image:** rays appear to diverge

## Key Insight

A converging lens forms a virtual image only when the object is *inside* the focal length.

A diverging lens *always* forms a virtual image for a real object.

# Outline of Part VI: Image Formation by Thin Lenses

Types of Lenses

Principal Rays and Image Types

The Thin Lens Equation

# The Thin Lens Equation

## Thin Lens Equation and Magnification — (cp2e\_ch25\_s6)

$$\boxed{\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}}$$

$$m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$$

### Variable Definitions

- ▶  $d_o$ : object distance ( $> 0$  for real objects)
- ▶  $d_i$ : image distance
- ▶  $f$ : focal length
- ▶  $m$ : lateral magnification

### Sign Conventions

- ▶  $d_i > 0$ : real image (far side)
- ▶  $d_i < 0$ : virtual image (same side)
- ▶  $f > 0$ : converging lens
- ▶  $f < 0$ : diverging lens
- ▶  $m > 0$ : upright;  $m < 0$ : inverted

# Lens Power and Image Summary

## Converging Lens

| Object          | Image                  |
|-----------------|------------------------|
| Beyond $2f$     | Real, inv., reduced    |
| At $2f$         | Real, inv., same size  |
| Btwn $f$ – $2f$ | Real, inv., enlarged   |
| Inside $f$      | Virtual, upright, enl. |

## Lens Power $P$

$$P = \frac{1}{f}$$

$f$  in m;  $P$  in diopters (D). *Eyeglass Rx in*

## Diverging Lens: Always

- ▶ Virtual
- ▶ Upright
- ▶ Reduced
- ▶ Same side as object

Virtual image cannot be projected.

Real image ( $d_i > 0$ ) can be projected.

## Example: Converging Lens Image — (cp2e\_ch25\_ex4)

### Image Location and Magnification

A converging lens has focal length  $f = 20.0$  cm. An object is placed  $d_o = 30.0$  cm from the lens. Find the image distance and magnification.

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$\frac{1}{d_i} = \frac{1}{20.0} - \frac{1}{30.0}$$

$$\frac{1}{d_i} = \frac{3 - 2}{60.0} = \frac{1}{60.0}$$

$$d_i = 60.0 \text{ cm}$$

$$m = -\frac{d_i}{d_o} = -\frac{60.0}{30.0} = -2.00$$

$d_i > 0$ : **real** image on far side.

$m < 0$ : **inverted**.

$|m| = 2$ : **enlarged** to twice the object size.

► Object is between  $f$  and  $2f$  (20 cm–40 cm). Consistent with real, inverted, enlarged? ✓

## Example: Magnifying Glass — (cp2e\_ch25\_ex5)

### Object Inside the Focal Length

A converging lens has  $f = 15.0$  cm. An object is placed  $d_o = 10.0$  cm from the lens. Find the image distance and magnification.

$$\frac{1}{d_i} = \frac{1}{15.0} - \frac{1}{10.0}$$

$$\frac{1}{d_i} = \frac{2 - 3}{30.0} = -\frac{1}{30.0}$$

$$d_i = -30.0 \text{ cm}$$

$$m = -\frac{-30.0}{10.0} = +3.00$$

$d_i < 0$ : **virtual** image on the same side as the object.

$m > 0$ : **upright**.

$|m| = 3$ : **enlarged** threefold.

This is exactly how a magnifying glass works!

► Why can't a real image form when  $d_o < f$ ?

## Check Your Understanding

A converging lens with  $f = 15$  cm has an object at  $d_o = 10$  cm. Is the image real or virtual? Upright or inverted? Larger or smaller than the object?

Come to lecture to see the answer.

## Check Your Understanding

A lens produces an image with magnification  $m = -3$ .  
What does this tell you about the image?

Come to lecture to see the answer.

## Part VII

# Image Formation by Mirrors

# Outline of Part VII: Image Formation by Mirrors

Types of Mirrors

The Mirror Equation

# Outline of Part VII: Image Formation by Mirrors

Types of Mirrors

The Mirror Equation

# Curved Mirrors

## Three Mirror Types

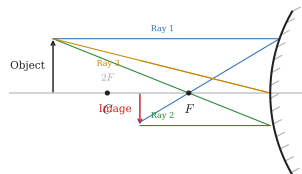
- ▶ **Plane:** flat,  $f = \infty$ , virtual upright image
- ▶ **Concave:** inward curve, converging,  $f > 0$
- ▶ **Convex:** outward curve, diverging,  $f < 0$

## Focal Length

$$f = \frac{R}{2}$$

$R$  = radius of curvature. Real images form

Concave Mirror: Principal Rays



# Outline of Part VII: Image Formation by Mirrors

Types of Mirrors

The Mirror Equation

# The Mirror Equation

## Mirror Equation — (cp2e\_ch25\_s7)

Identical in form to the thin lens equation:

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} = \frac{2}{R}$$

Magnification:  $m = h_i/h_o = -d_i/d_o$

## Sign Conventions for Mirrors

- ▶  $d_i > 0$ : real image, *in front* of mirror
- ▶  $d_i < 0$ : virtual image, *behind* mirror
- ▶  $f > 0$ : concave (converging)
- ▶  $f < 0$ : convex (diverging)

## Lens vs. Mirror

|            | Lens     | Mirror     |
|------------|----------|------------|
| Mechanism  | Refract. | Reflect.   |
| Converging | $f > 0$  | $f > 0$    |
| Real image | Far side | Front face |
| Diverging  | $f < 0$  | $f < 0$    |

# Concave Mirror Image Summary

## Concave Mirror

| Object          | Image                  |
|-----------------|------------------------|
| Beyond $2f$     | Real, inv., reduced    |
| At $2f$         | Real, inv., same size  |
| Btwn $f$ – $2f$ | Real, inv., enlarged   |
| At $f$          | No image               |
| Inside $f$      | Virtual, upright, enl. |

## Applications

**Concave:** makeup mirrors, headlights

**Convex:** security, rear-view mirrors

## Convex Mirror: Always

- ▶ Virtual
- ▶ Upright
- ▶ Reduced

Same pattern as a diverging lens.

## Example: Concave Mirror — (cp2e\_ch25\_ex6)

### Image in a Concave Mirror

A concave mirror has radius of curvature  $R = 60.0$  cm. An object is placed  $d_o = 90.0$  cm in front. Find the image distance and magnification.

$$f = \frac{R}{2} = 30.0 \text{ cm}$$

$$\frac{1}{d_i} = \frac{1}{30.0} - \frac{1}{90.0}$$

$$\frac{1}{d_i} = \frac{3 - 1}{90.0} = \frac{2}{90.0}$$

$$d_i = 45.0 \text{ cm}$$

$$m = -\frac{45.0}{90.0} = -0.500$$

$d_i > 0$ : **real** image in front of mirror.

$m < 0$ : **inverted**.

$|m| = 0.5$ : **reduced** to half size.

Object is beyond  $2f$  ( $90 \text{ cm} > 60 \text{ cm}$ ) — consistent with real, inverted, reduced.

✓

► If a bulb is placed at  $f = 30$  cm, where does the image form?

## Example: Convex Security Mirror — (cp2e\_ch25\_ex7)

### Convex Mirror

A convex security mirror has focal length  $f = -0.500$  m. An object is  $d_o = 3.00$  m from the mirror. Find the image distance and magnification.

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$\frac{1}{d_i} = \frac{1}{-0.500} - \frac{1}{3.00}$$

$$\frac{1}{d_i} = -2.00 - 0.333 = -2.333$$

$$d_i \approx -0.429 \text{ m}$$

$$m = -\frac{-0.429}{3.00} \approx +0.143$$

$d_i < 0$ : **virtual** image behind mirror.

$m > 0$ : **upright**.

$|m| < 1$ : **reduced**.

Wide field of view — useful for security and car mirrors.

► Why do convex mirrors always give upright, reduced images for real objects?

## Check Your Understanding

The back of a spoon acts like a convex mirror. What type of image do you expect to see when you look at your reflection?

Come to lecture to see the answer.

## Check Your Understanding

A makeup (concave) mirror with  $f = 30$  cm is used with your face at  $d_o = 20$  cm. What kind of image forms, and why is this useful?

Come to lecture to see the answer.

## Check Your Understanding

Why is the bulb in an automobile headlight placed at the focal point of a concave mirror reflector?

Come to lecture to see the answer.